

The solution we propose addresses the lack of pediatric clinical trials in Canada by establishing a federal fund and incentives for researchers and institutions conducting pediatric research. Currently, drug development for children is far behind adult research largely because trials involving children are much more expensive, complex, and less commercially attractive than adult trials, due to smaller patient populations and greater ethical concerns (Health Canada, 2022). Therefore, by reducing the financial burden associated with pediatric clinical trials, this solution aims to help both public and private sponsors toward greater investments in this underserved area.

Concretely, the solution would combine two mechanisms. First, the program would establish a dedicated pediatric research grant stream through an existing organization such as the Canadian Institutes of Health Research (CIHR). This funding would support researcher-led pediatric clinical trials, specialized trial resources such as age-appropriate consent procedures and medication formulations, and collaboration among children's hospitals across Canada. Second it would introduce tax-based incentives for sponsors, similar to the Scientific Research and Experimental Development (SR&ED) tax credit (Canada Revenue Agency [CRA], 2026), along with longer data protection or market exclusivity for medications that complete pediatric trials and receive approval for use in children. Together, these levers would lower both the upfront cost and the long-term commercial risk of pursuing pediatric research.

By including direct public investments with financial incentives, this approach is designed to engage both academic researchers and drug manufacturers, ensuring that funding gaps are addressed from multiple angles rather than relying on a single lever. It also allows Canada to keep up with jurisdictions such as the United States and European Union, which have already implemented similar pediatric incentive frameworks, such as the Best Pharmaceuticals for Children Act and Pediatric Regulation (European Commission, n.d.; U.S. Food and Drug Administration [FDA], 2018), reducing the risk that Canadian children are left dependent on off-label or adult dosing data due to a lack of domestic pediatric evidence (Health Canada, 2024).

This proposed solution uses three government levers in order to achieve the desired outcome. These levers include; grant and contribution programs, regulations and data collection and reporting. Together, these levers would provide funding for pediatric research and encourage pharmaceutical companies to perform more pediatric trials and to ensure the found evidence is available to the public. The primary policy lever would be a grant program administered through a dedicated pediatric research stream at the Canadian Institutes of Health Research (CIHR). Pediatric trials are often more expensive and difficult to organize than adults' trials because they require specialized researchers working with increased safeguards and different age groups each with their unique needs and cooperation among multiple hospitals. Federal grants would help cover these costs and reduce the financial risk faced by researchers and institutions. The main portion of the funding would go towards supporting trial coordinators, pediatric consent and assent procedures, pharmacokinetic studies, and the development of age-appropriate formulations. Priority could also be given to trials involving neonates, rare diseases, and other pediatric populations for which treatment evidence is especially limited.

The second policy lever would be government regulations. This lever would be used to encourage pharmaceutical companies to complete pediatric studies and apply for approval to use their medications in children. Canada currently provides innovative drugs with eight years of

data protection and allows an additional six months when acceptable pediatric trial information is submitted (Health Canada, 2025). The proposed program would build upon this approach by offering a longer period of protection when a company completes the required pediatric trials, receives approval for pediatric use, and provides appropriate information about dosing, safety, and formulation. This would give manufacturers a clearer reason to invest in pediatric research despite the smaller market for children's medications.

The final lever would focus on data collection and reporting. Researchers and companies receiving grants or regulatory incentives would be required to register their pediatric trials, report results for each pediatric age group, and publish their findings within a set period. These requirements would allow CIHR and Health Canada to measure the program's progress and determine whether government support is producing meaningful improvements in pediatric medication access. Mandatory reporting would also reduce the number of completed pediatric studies that remain unpublished and prevent valuable evidence from being lost.

The solution addresses the financial, practical, and commercial barriers that limit pediatric drug development. Pediatric trials are costly and difficult to conduct because they involve smaller patient populations, different age groups, complex recruitment, and additional ethical requirements (Caldwell et al., 2004; Lagler et al., 2021). The proposed solution would help researchers manage these challenges, while regulatory incentives would encourage pharmaceutical companies to invest in the smaller pediatric market. Finally, mandatory data collection and reporting would reduce the loss of evidence caused by delayed or unpublished trial results (Nebeska et al., 2025). Together, these levers could increase the availability of approved pediatric dosing information and age-appropriate formulations, reducing unnecessary reliance on off-label medications.

Increasing funding for pediatric clinical trials directly aligns with Health Canada's mandate to protect and improve the health of Canadians, including children. By investing dedicated federal resources into pediatric research, Health Canada strengthens its core responsibility of ensuring that therapeutic products are rigorously evaluated for the specific age groups that rely on them (Health Infobase, 2025). Expanded funding would allow researchers to replace unsafe or poorly studied medications with treatments supported by strong pediatric safety and efficacy data. This financial support also acts as a catalyst for pharmaceutical companies, helping them overcome long-standing economic barriers and encouraging greater investment in pediatric drug development. As noted by Raja, P., et al. (2020), targeted funding can accelerate innovation while also improving alignment between Canadian regulatory practices and international agencies such as the FDA and EMA.

This solution also fits Health Canada's broader goals of improving access to pediatric medicines, increasing the development of age-appropriate formulations, and expanding the availability of reliable pediatric data for Canadians. Additional funding would support the implementation of pediatric development plans, which Health Canada now requires for new drug submissions. Strengthening these plans ensures that pediatric considerations are integrated early in the drug development process rather than delayed until after adult studies are completed (Government of Canada, 2023).

Moreover, increased funding would allow Health Canada to expand clinical trial frameworks across the country, supporting collaboration among children's hospitals and specialized research units. This investment also reinforces Health Canada's commitment to health equity by ensuring that Canadian children are not excluded from modern treatments and emerging therapies simply because pediatric trials are too costly or complex to conduct. As Hepburn, CM., et al. (2023) emphasize, equitable access to safe and effective medicines is a fundamental component of public health. By reducing financial barriers and supporting robust pediatric research infrastructure, Health Canada can ensure that children receive therapies that are scientifically validated, age-appropriate, and responsive to their unique developmental needs.

There are different people of interest involved in this solution, one group being researchers. Several potential age-appropriate drugs do not get properly tested due to the lack of funding for research; therefore, increasing the funding and incentives for pediatric clinical trials would allow researchers to study the efficacy and safety of different medications in their target population. Those who are interested in this field would appreciate the financial aid; however, it would increase the workload of ethics committees, which would have to revisit the rules regarding the involvement of children in research. The increased incentives and attention for pediatric clinical trials would also attract pharmaceutical companies, which would provide more support and reach for researchers.

Increased funding would require the involvement of the federal government, which would have to approve this solution and ensure that research funding agencies, such as CIHR, are provided with the funds required to develop a grant specific to pediatric studies. The reaction of government officials will depend on the country's priorities, considering the current state of foreign affairs and focus on other important issues. When considering the reaction of the members of government, it is also important to consider the reaction of the citizens, as it would be their tax dollars that the government would provide to the research funding agencies. Considering how many Canadians struggle to earn enough for necessities such as food and shelter, it may not be easy for the government to directly increase taxes. However, a negative reaction from the public may be reduced if tax money that is used for less important reasons is redirected to fund pediatric clinical trials.

Finally, pediatric patients are the main group involved, as increased funding would allow for more age-appropriate medications to be developed for various illnesses. The main driver behind research is funding; therefore, such a solution would provide hope to the patient and their families. Although some families may be wary of allowing their child to participate in a clinical trial, if there are no other alternatives, it will most likely be a welcome opportunity for their child to battle their illness.

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